

VANSH BADJATE

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SUMMARY

AI/ML engineering student with hands-on experience building Generative AI applications using LLMs, Vision-Language Models, RAG, and FastAPI. Shipped production invoice-extraction and multilingual legal-assistant pipelines by fine-tuning and prompt-engineering Qwen2.5-VL, and evaluated LLM response quality across post-training workflows during three AI internships. Built five end-to-end AI and full-stack projects, from RAG pipelines to a deployed SQL-copilot tool, for real-world use cases.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java

Generative AI & ML: LLMs, RAG (Retrieval-Augmented Generation), Vision-Language Models (VLMs), LoRA Fine-Tuning, Prompt Engineering, Agentic AI, Computer Vision (YOLOv8, OpenCV)

Frameworks & Libraries: FastAPI, LangChain, Transformers, Node.js, Express.js, Flask, React.js, Scikit-learn, TensorFlow

Databases: ChromaDB (Vector DB), MongoDB, PostgreSQL, MySQL, SQLite

Tools & Platforms: Ollama, Docker, Git, GitHub, Redis, BullMQ

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

EXPERIENCE

AI Development Intern - Sai Computech, Kopergaon

May 2026 - Present

- Built a production invoice-extraction pipeline validated on 300+ real Indian GST invoices, by piloting LoRA fine-tuning on Qwen2.5-VL and then shipping a higher-accuracy prompt-engineered Qwen2.5-VL-7B pipeline with post-processing after comparing both approaches.
- Improved invoice-upload UX by removing single-block wait times, by building a Server-Sent Events (SSE) streaming API that returns extracted invoice fields section-by-section as they're generated instead of one delayed response.
- Shipped 10 production REST API endpoints for invoice upload, extraction, status tracking, retrieval, and deletion, by building Node.js/Express services backed by a BullMQ/Redis job queue and PostgreSQL, with user-scoped access control.
- Enabled data-driven model selection for the team, by building internal batch-testing tools that ran accuracy reports across invoice datasets and alternative OCR/vision-language model candidates.

LLM Post-Training Intern - Ethara AI, Gurugram (Remote)

Jan 2026 - May 2026

- Sustained a working target of 80+ LLM-response evaluations per day against reasoning-quality, factual-accuracy, and instruction-adherence rubrics, by running structured post-training evaluation workflows.
- Surfaced recurring model failure patterns for the training team, by analyzing model outputs for hallucinations and reasoning gaps and documenting findings to support benchmarking.
- Maintained evaluation quality across diverse multimodal tasks, by conducting prompt evaluation, response validation, and multimodal assessment against team productivity targets.

PROJECTS

NyayAI | Multilingual AI Legal Assistant

GitHub: github.com/vanshbadjate07/NyayAI-AI-Legal-Assistant-for-Indian-Law

- Delivered source-cited legal answers across 5 Indian legal domains (RTI, Tenancy, Family, Consumer, Criminal Law) in English, Hindi, and Marathi, by building a multilingual RAG pipeline with BAAI/bge-m3 embeddings and ChromaDB retrieval over official Indian Acts.
- Enabled document-based legal queries from photographed notices and agreements, by integrating OCR (EasyOCR + Moondream2) with a self-hosted Qwen2.5:3B (Ollama) generation layer; secured Government of India copyright registration (SW-2026022947).

Tech Stack: Python, FastAPI, ChromaDB, Ollama (Qwen2.5:3B), BAAI/bge-m3, EasyOCR, SQLite, React

QueryMind | AI SQL/DBMS Assistant

GitHub: github.com/vanshbadjate07/QueryMind

- Shipped a deployed AI chat assistant that writes, fixes, and optimizes SQL and answers natural-language questions over uploaded CSV/Excel datasets, by building an auto-mode-detecting FastAPI backend with SQLAlchemy and JWT authentication.
- Prevented unsafe or runaway queries, by validating every LLM-generated SQL statement with sqlglot (read-only, single-statement, row-limited) before execution, and removed single-key rate-limit failures via multi-key Groq API rotation with automatic model fallback.

Tech Stack: Python, FastAPI, React, MySQL, SQLAlchemy, Groq API, sqlglot, JWT

SafeVision | Real-Time AI Surveillance System

GitHub: github.com/vanshbadjate07/SafeVision

- Won Gold Medal at AI Innovation Expo 2026 (IBM CareerTiQ), by building a real-time surveillance system combining zone-based intrusion detection, fine-tuned YOLOv8 weapon detection, mask detection, and face-recognition access control across a 6-step per-frame pipeline.
- Delivered evidence-backed alerting, by building a stateful Twilio WhatsApp bot that sends detection snapshots with adaptive frame-skipping (6-10 frames) and a live MJPEG dashboard for remote monitoring and control.

Tech Stack: Python, OpenCV, YOLOv8, Flask, dlib / face_recognition, Twilio

ReasonEval | LLM Reasoning Quality Evaluation Engine

GitHub: github.com/vanshbadjate07/LLM-Reasoning-Quality-Evaluation-Engine

- Built a rule-based reasoning evaluator that scores LLM outputs 0-100% across 4 dimensions (logical consistency, completeness, instruction-following, hallucination risk), by encoding 20+ heuristic checks into a First-Principles grading engine.
- Delivered a self-hosted, zero-API-cost evaluation tool, by pairing a Streamlit scoring dashboard with a Flask REST API on local Ollama models (Llama 3.2/3.1), forcing structured reasoning before every grade.

Tech Stack: Python, Flask, Streamlit, Ollama (Llama 3.2 / 3.1), Pandas

EDUCATION

Sanjivani University, Kopergaon B.Tech in Computer Science (AI & ML) CGPA: 8.50/10	2024 - 2027
Sanjivani K.B.P. Polytechnic, Kopergaon Diploma in Computer Technology 84.51 %	2021 - 2024

CERTIFICATIONS

- **Agentic AI Professional Certificate** - LinkedIn Learning
- **Building AI Agents with AutoGen** - Microsoft
- **Build AI Agents with LangGraph** - LinkedIn Learning
- **Introduction to Prompt Engineering for Generative AI** - LinkedIn Learning
- **Introducing Generative AI with AWS** - AWS / Udacity

INTELLECTUAL PROPERTY

NyayAI: AI-Based Legal Document Analysis and Guidance System - Copyright Registered, Government of India
Application No. SW-45302/2025-CO | Certificate No. SW-2026022947 | Registered: 29 May 2026 | Co-author: Vaishnavi Nitin Nikumbh

ACHIEVEMENTS & ACTIVITIES

- Gold Medal, AI Innovation Expo 2026 (IBM CareerTiQ) - SafeVision: Real-Time AI Surveillance System
- Grand Finalist (Top 25/60 Teams), Kumbha Visitor Experience Innovation Challenge 2026
- Vice President, AIML Department Student Association (ADSA), Sanjivani University (2024-2025)
- Participant, SUNHACKS 2025 National Hackathon, Sandip University